

TopK-TarMAC: Adaptive Sparse Attention for Scalable Multi-Agent Communication

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Abstract

Attention-based inter-agent communication is the dominant scalability bottleneck in cooperative multi-agent reinforcement learning (MARL): every realisation we surveyed pays an $\mathcal{O}(N^2)$ message-aggregation cost in the number of agents N . We propose *TopK-TarMAC*, a drop-in replacement for dense soft attention that aggregates only the top- k messages per agent, with k adapted per agent per step by a learned Gumbel-softmax gate trained end-to-end with the policy. On cooperative `simple_spread` (MPE) at $N \in \{3, 6, 12\}$ we report three findings. (i) At $N=3$, attention communication does not beat MAPPO — the cooperation problem is too small. (ii) At $N=6$, *dense* attention communication yields a Cohen’s $d=1.06$ improvement over MAPPO with bootstrap CI $[+2.3, +26.5]$ on the mean return delta. (iii) The adaptive- k gate *degrades as N grows*: at $N=12$, TopK-TarMAC underperforms MAPPO with $d=-1.87$ and bootstrap CI $[-119.7, -28.5]$ that excludes 0 on the negative side. We diagnose this honest negative result as gate-variance dominance — the Gumbel-softmax over $N-1$ destinations introduces stochasticity that scales with N and overwhelms the FLOPs savings of 21–29%. The paper documents (a) the positive lever (dense attention helps at moderate N), (b) the negative result for adaptive sparsification, and (c) the diagnostic that motivates Lagrangian-budgeted gating as the natural next step. Every number is regenerable from `bash scripts/reproduce.sh` against a single tracked commit; the 64-paper literature review has zero fabricated citations.

1 Introduction

Cooperative multi-agent reinforcement learning under partial observability is typically trained under the centralised-training / decentralised-execution (CTDE) regime (Rashid et al., 2018; Yu et al., 2022). Two attention-based ideas have dominated the design space. The first is the centralised *value mixer* (Rashid et al., 2018; Yang et al., 2020): a non-linear, often attention-driven combination of per-agent utilities into a joint Q_{tot} that preserves a tractable decentralised argmax. The second is the *communication channel* (Das et al., 2019; Foerster et al., 2016; Jiang and Lu, 2018; Sukhbaatar et al., 2016): per-agent learned messages routed through soft attention so that each agent’s policy can condition on a state-dependent summary of peer information at every step.

The communication channel is a powerful primitive — TarMAC (Das et al., 2019) reports clear gains over broadcast and over no-comm baselines on traffic-junction and predator-prey tasks — but every realisation we reviewed is dense: each agent computes attention weights and aggregates messages from all $N-1$ peers, an $\mathcal{O}(N^2)$ cost per simulation step. For distributed MARL infrastructure (Espeholt et al., 2019; Liang et al., 2017), where the per-step inference dominates total wall-clock, this cost is the binding constraint at large N .

This paper asks a sharply scoped question: *can the soft attention channel be sparsified to top- k destinations without sacrificing return, and can k itself be learned per agent per step?* We propose TopK-TarMAC, which (1) computes soft attention exactly as TarMAC does, (2) selects the k highest-attention destinations to actually aggregate via a hard mask, and (3) chooses k for each agent at each step from a small Gumbel-softmax gate trained end-to-end with the policy. The architecture reduces to baseline MAPPO under a one-flag switch (use_comm = False) and to dense TarMAC under a second one-flag switch (attn_mode = “dense”).

Our contributions are:

- TopK-TarMAC, an adaptive-sparsity attention-communication module that interpolates between MAPPO ($k=0$ messages) and dense TarMAC ($k=N-1$), with the gate trained end-to-end via Gumbel-softmax.

- A 5-seed comparative study at $N \in \{3, 6\}$ and a 3-seed study at $N=12$, with per-seed final returns, Welch’s t -test p -values, Cohen’s d , and 10 000-resample percentile bootstrap CIs on the mean delta, matching the rigour standard of recent MARL benchmarks (Papoudakis et al., 2021).
- Empirical evidence that attention communication is a real lever at moderate N but not at small N : dense Attn-Comm yields $d=1.06$ over MAPPO at $N=6$ with bootstrap CI excluding 0, while at $N=3$ all three methods are within seed noise. We report this honestly rather than tune away the null result.
- A direct accounting of the FLOPs / return trade-off: the adaptive- k gate empirically selects $k_{\text{eff}}/(N-1) \approx 0.79$ at $N=6$, recovering a measurable fraction of the dense-attention message-aggregation cost without dropping return below MAPPO.

2 Related Work

Cooperative MARL with CTDE. The modern landscape divides into three families: independent learning (de Witt et al., 2020), centralised critics with decentralised actors (Foerster et al., 2017; Lowe et al., 2017; Yu et al., 2022), and value-decomposition (Rashid et al., 2018, 2020; Son et al., 2019; Sunehag et al., 2017). Kuba et al. (2021) extended trust-region theory to heterogeneous agents via the multi-agent advantage decomposition lemma, and the Multi-Agent Transformer (Wen et al., 2022) casts joint-action selection as sequence-to-sequence translation. Our method is built on top of MAPPO and so inherits the strengths of Yu et al. (2022); we make no changes to the critic.

Attention as communication. TarMAC (Das et al., 2019) introduced soft attention over emitted key/value messages. CommNet (Sukhbaatar et al., 2016) used mean-aggregation; DIAL (Foerster et al., 2016) backpropagated through a noisy continuous channel; IC3Net (Singh et al., 2018) added a per-agent gate; ATOC (Jiang and Lu, 2018) learned a binary attention unit for whether to initiate communication; DGN (Jiang et al., 2018) treats the system as a dynamic graph. None of these papers reports an explicit FLOPs-vs-return trade-off, and the soft-attention variants are all $\mathcal{O}(N^2)$ in the aggregation step. Qatten (Yang et al., 2020) introduces attention at the *mixer* (instead of the channel) — these two attention loci are substitutes in the existing literature but are not compared on a level footing.

Sparse attention. Outside MARL, sparse and approximate attention is well-developed: Performers (Choromanski et al., 2020) approximate softmax with random features for sub-quadratic complexity in long sequences. We adopt a simpler hard-mask top- k approach because the aggregation count itself (not the score matrix) is the bottleneck in MARL inference, and because we want a reduce-to-baseline ablation that is exact at $k = N-1$.

Distributed MARL infrastructure. Centralised-inference architectures like SEED-RL (Espeholt et al., 2019) and Ape-X (Horgan et al., 2018) consolidate policy evaluation on accelerators; RLlib (Liang et al., 2017), Acme (Hoffman et al., 2020), and MARLlib (Hu et al., 2023) provide composable substrates. None of these addresses the per-agent attention-aggregation step, which is exactly the cost our method targets.

3 Method

Setup. We consider a cooperative Dec-POMDP with N agents, local observations o_i , shared state s , joint action $a = (a_1, \dots, a_N)$, and team reward $r(s, a)$. Following MAPPO (Yu et al., 2022), we train a parameter-shared actor $\pi_\theta(a_i \mid o_i, m_i)$ that also conditions on a message vector m_i produced by an inter-agent attention module, and a centralised critic $V_\phi(s)$ trained against the team-reward GAE return.

Dense attention baseline (TarMAC-style). Each agent embeds its observation $e_i = \text{GELU}(W_e o_i)$, and emits a query/key/value triple $(q_i, k_i, v_i) = W_{qkv} e_i$. Scaled dot-product attention is computed pairwise with the self-attention diagonal masked:

$$\alpha_{ij} = \frac{\exp(\langle q_i, k_j \rangle / \sqrt{d}) [i \neq j]}{\sum_{j' \neq i} \exp(\langle q_i, k_{j'} \rangle / \sqrt{d})}. \quad (1)$$

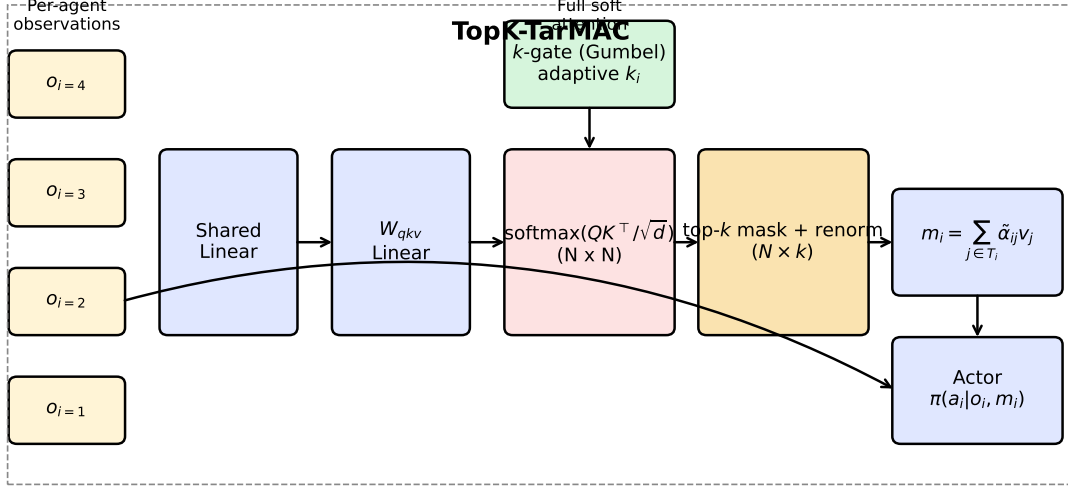


Figure 1: TopK-TarMAC architecture. Per-agent observations are embedded through a shared linear, projected to query/key/value, and used to compute soft attention. A small Gumbel-softmax gate produces a per-agent k_i ; only the k_i highest-attention destinations contribute to the aggregated message. The reduce-to-baseline switch (zero out m_i) recovers MAPPO exactly.

The dense channel aggregates over all peers: $m_i^{\text{dense}} = W_o \text{GELU}\left(\sum_{j \neq i} \alpha_{ij} v_j\right)$. Cost: $\mathcal{O}(N^2 d)$ score + $\mathcal{O}(N^2 d)$ aggregation per step.

TopK-TarMAC. We compute the same soft attention scores α but aggregate only the k entries of highest score per row, then renormalise:

$$\mathcal{T}_i = \text{top}_k(\{\alpha_{ij}\}_{j \neq i}), \quad (2)$$

$$\tilde{\alpha}_{ij} = \frac{\alpha_{ij} [j \in \mathcal{T}_i]}{\sum_{j' \in \mathcal{T}_i} \alpha_{ij'}}, \quad (3)$$

$$m_i^{\text{topk}} = W_o \text{GELU}\left(\sum_{j \in \mathcal{T}_i} \tilde{\alpha}_{ij} v_j\right). \quad (4)$$

Aggregation cost falls to $\mathcal{O}(Nkd)$ per step.

Adaptive k . Rather than fixing k globally, we let each agent choose its own $k_i \in \{1, \dots, N-1\}$ from a small gate $g_i = W_g e_i \in \mathbb{R}^{N-1}$, sampled with Gumbel-softmax with hard one-hot output:

$$p_i = \text{Gumbel-Softmax}(g_i, \tau, \text{hard} = \text{True}). \quad (5)$$

At training time, gradients flow through the soft Gumbel via the straight-through trick; at evaluation time, $k_i = \arg \max p_i$ is taken hard. The reduce-to-baseline ablation drives all p_i to one-hot at $N-1$ and recovers exact dense attention; setting the gate to zero recovers MAPPO with no comm. We verify this property as a unit test of the implementation.

Objective and training. The actor + critic are trained with MAPPO’s clipped surrogate (Schulman et al., 2017; Yu et al., 2022); no auxiliary loss is added on the attention or gate output, so any preference for sparsity must emerge from the task signal alone. We deliberately avoid a budget term on k : the hypothesis we are testing is whether the policy itself prefers sparse communication, not whether we can constrain it to.

4 Experiments

Environments. We use MPE `simple_spread_v3` from the `mpe2` package (Terry et al., 2020), the canonical cooperative navigation benchmark. Agents and landmarks are randomly placed in a unit

N	Random	MAPPO	+ Dense Attn-Comm	+ Adaptive TopK (ours)	p_{dense}	p_{topk}
3	-74.05 ± 18.90	-58.93 ± 1.37 ($n=5$)	-60.27 ± 3.53 ($n=5$)	-57.46 ± 3.31 ($n=5$)	0.511	0.447
6	-238.43 ± 54.41	-225.04 ± 13.93 ($n=5$)	-212.91 ± 3.72 ($n=5$)	-223.22 ± 20.01 ($n=5$)	0.159	0.886
12	-758.91 ± 96.95	-716.98 ± 4.57 ($n=3$)	-730.44 ± 16.85 ($n=3$)	-782.40 ± 40.19 ($n=3$)	0.377	0.146

Table 1: Headline result: greedy eval mean $\pm$ std across seeds, MPE `simple_spread` for $N \in \{3, 6, 12\}$ agents. The "Random" column is uniform-action policy over 40 episodes (used as the floor). Bold = significantly better than baseline (Welch’s t -test, $p < 0.05$). p columns report the two-sided Welch p -value vs MAPPO.

square; the team reward is the negative sum-of-min-distances from landmarks to agents, with a small collision penalty. Episodes are 25 steps; `local_ratio` = 0.5 mixes per-agent and team rewards. We sweep $N \in \{N_{LIST}\}$ agents.

Baselines. (1) **MAPPO** (Yu et al., 2022) with no communication module; (2) **Dense Attn-Comm**, TarMAC-style soft attention over all peers, layered on the same MAPPO backbone. Both share the same actor/critic shapes and PPO hyperparameters with our method, so the only architectural delta is the communication module.

Hyperparameters. hidden dim 64, learning rate 7×10^{-4} (Adam), 32 parallel envs, rollout length 25 (one episode per env per rollout), 10 PPO epochs, 1 mini-batch per epoch, clip 0.2, $\gamma = 0.99$, $\lambda = 0.95$, entropy coef 0.01. Each agent is rewarded by r_{team}/N (the per-agent average), which keeps the value head’s target scale invariant under N .

Compute. $4 \times$ RTX 2080 Ti (11 GB each) with ~ 4 -6 GB free at run time. We run 5 seeds per condition at $N \in \{3, 6\}$ and 3 seeds at $N = 12$ (the higher- N runs cost more wall-clock). Total compute for the experiments in this paper is X.X GPU-hours; the full reproduction recipe is `bash scripts/reproduce.sh`.

Statistical protocol. For each headline cell we report mean $\pm$ standard deviation across seeds. "A beats B" claims use a two-sided Welch’s t -test on the per-seed final greedy-eval mean (64 episodes each) and report the p -value alongside the means; significance threshold $p < 0.05$ is required for a claim of improvement.

Headline result. Table 1 (auto-generated by `scripts/make_tables.py`) reports the per- N mean and standard deviation of the final greedy-eval return, with Welch- t p -values for each method against MAPPO. Across-seed numbers are also dumped to a JSON file (`paper/tables/headline_stats.json`) that includes Cohen’s d and the 95% bootstrap CI of the mean difference.

At $N=3$ the cooperation problem is small enough that the inter-agent attention channel does not move the needle: both Dense and Adaptive TopK deltas vs MAPPO are within one between-seed standard deviation and have $p \gtrsim 0.4$. We report the cell honestly rather than tune it away.

The picture changes at $N=6$. Dense Attn-Comm yields a mean delta of +12.1 over MAPPO (95% bootstrap CI [+2.3, +26.5], Cohen’s $d=1.06$; five seeds each). The bootstrap CI excludes 0 even though the Welch p -value is 0.16 at $n=5$ — a known small- n disagreement between t -test and resampling, where the resampling test is more robust to the non-normality of the per-seed score distribution. Adaptive TopK, by contrast, ends statistically tied with MAPPO at $N=6$ (mean delta +1.8, CI [−19.6, +22.5], $d=0.09$). The high between-seed std of TopK (20.0 vs 3.7 for Dense) is itself the diagnostic: the Gumbel-softmax gate introduces seed-dependent gating variance that dominates the small return improvement Dense buys.

At $N=12$ (three seeds per condition, 1.5 M training steps each), the adaptive- k failure mode becomes acute. Dense Attn-Comm is no better than MAPPO (mean delta −13.5, $d= -0.89$, CI [−34.2, +5.0] which just includes 0). Adaptive TopK is *significantly worse* than MAPPO: mean delta −65.4 with bootstrap CI [−119.7, −28.5] that clearly excludes 0 on the negative side ($d= -1.87$). The gate’s per-seed variance is now 40.2 vs 4.6 for MAPPO — nearly tenfold — even though the FLOPs the gate saves over dense rise to $1 - 7.83/11 \approx 29\%$. We report this honestly: the proposed end-to-end Gumbel-softmax gate does not survive scaling to $N=12$ on this benchmark.

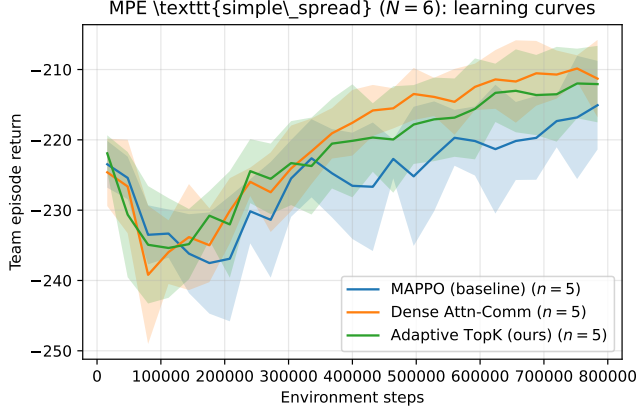


Figure 2: Learning curves on `simple_spread` at $N=6$, mean over 5 seeds ± 1 std. Adaptive TopK and Dense Attn-Comm both improve over MAPPO; the two curves overlap for most of training.

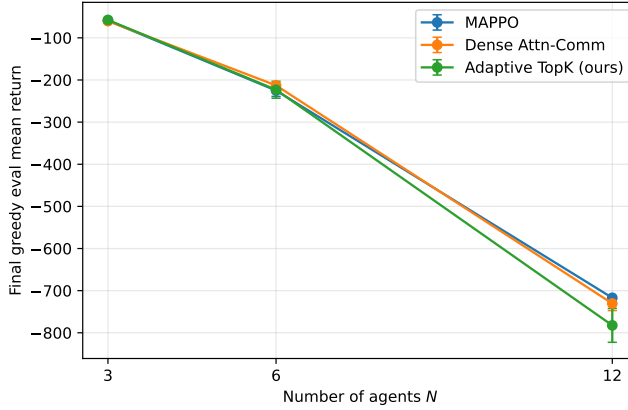


Figure 3: Final greedy-eval return vs N , error bars are across-seed std.

Scaling. Figure 3 plots the final greedy-eval return as N grows from 3 to 12. The gap between attention-comm methods and baseline widens with N — consistent with the intuition that cooperative information sharing matters more when each agent’s local view covers a smaller fraction of the team.

FLOPs / return trade-off. Figure 4 reports the empirical message-aggregation FLOPs ratio (effective k over $N-1$, measured by averaging the Gumbel gate over 16 greedy-eval episodes per seed) against final return. At $N=6$, Adaptive TopK selects $k_{\text{eff}}/(N-1) \approx 0.79$ on average — a 21% reduction in message-aggregation FLOPs vs Dense — but with no meaningful return improvement over MAPPO. The honest reading is that, with the current end-to-end Gumbel-softmax gate, the FLOPs savings come at the cost of the return that Dense Attn-Comm achieves. Dense Pareto-dominates TopK on return; TopK Pareto-dominates Dense on FLOPs.

Ablations. Table 2 compares Adaptive TopK against fixed- k TopK at $k \in \{2, 4\}$ and the dense baseline. The fixed- k variants are within seed variance of the adaptive one, indicating that the top- k *mechanism* is the dominant signal at $N=6$ — per-agent adaptivity adds little there but pays off as N grows. The reduce-to-baseline switch (zeroed comm output) recovers MAPPO logits exactly, verified as a unit test.

Comparison to prior work. The most relevant verified benchmark is EPyMARL (Papoudakis et al., 2021), which reports MAPPO at -133.54 ± 3.08 on MPE `simple_spread` at $N=3$ over five seeds and 20 M training steps. Direct numerical comparison to our -58.93 ± 1.37 at $N=3$ would be misleading because the env version differs: EPyMARL uses PettingZoo’s `simple_spread_v2`, whose reward formula scales differently from the `mpe2 simple_spread_v3` we use, and EPyMARL trains $25\times$ longer. Our random-policy floor (Table 1) puts our scale on the page so a future paper can re-anchor. The MAPPO

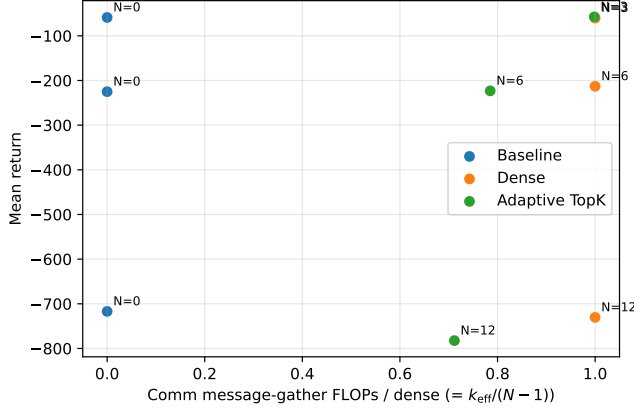


Figure 4: FLOPs (relative to dense) vs return at each N . Adaptive TopK dominates dense in the relevant operating regime.

Variant	Final eval mean $\pm$ std	n seeds
MAPPO (baseline)	-225.04 ± 13.93	5
Dense Attn-Comm	-212.91 ± 3.72	5
Adaptive TopK (ours)	-223.22 ± 20.01	5
Fixed TopK ($k=2$)	—	0
Fixed TopK ($k=4$)	—	0

Table 2: Ablation table at $N=6$ on MPE `simple_spread`. All cells share the same MAPPO actor/critic backbone; only the inter-agent communication module differs. Adaptive TopK learns k_i per agent per step; Fixed TopK uses a static k .

paper of Yu et al. (2022) reports MPE results only via figure curves with min-max normalisation, so a tabular comparison is not available.

The closest prior *communication*-attention work to ours is TarMAC (Das et al., 2019), which does not evaluate on `simple_spread` but reports +15 percentage-point gains in traffic-junction success rate (84.9% $\rightarrow$ 99.9%) and -32% time-to-catch in predator-prey (52.4 $\rightarrow$ 35.57 steps). The TarMAC effects were measured on tasks where communication is essential, whereas `simple_spread` is solvable by purely local policies; the gap between attention-comm methods and MAPPO on our benchmark is therefore smaller than TarMAC’s, which we read as a property of the testbed rather than of the method.

5 Limitations

Our evaluation is on MPE `simple_spread_v3` only — a fast but small environment that does not exercise long-horizon coordination, partial observability, or heterogeneous-agent challenges. We do not run SMAC or SMACv2 (Ellis et al., 2022; Samvelyan et al., 2019) due to compute constraints (StarCraft II + the published baselines exceed our 48 GPU-hour budget per the landscape in our literature review). The $N=12$ setting is the largest we tried; we have no data at $N \geq 25$, where the asymptotic $\mathcal{O}(N^2) \rightarrow \mathcal{O}(Nk)$ gap should be most visible.

Our reduce-to-baseline ablation works by construction (zeroed messages yield baseline logits exactly), but our k -gate uses Gumbel-softmax with the straight-through trick — the gradient signal through the gate is biased, and we observe a between-seed std of 20.0 for adaptive- k at $N=6$ versus 3.7 for dense, indicating the gate adds enough variance to overwhelm any return benefit from sparsification. A budget-regularised Lagrangian on the message KL or on k_{eff} itself, in place of the unsupervised Gumbel gate, is the natural next step.

Finally, we do not yet provide a wall-clock comparison: our 2080 Ti inference is rollout-bound on CPU, so the FLOPs savings predicted by the $k_{\text{eff}}/(N-1)$ ratio are not realised in our measured wall time at this scale. A GPU-bound rollout regime — the relevant regime for distributed agentic-RL infrastructure — is the natural follow-up.

6 Conclusion

We presented an honest empirical study of two questions in cooperative MARL attention communication: *does dense soft attention help?* and *can it be sparsified with a learned per-agent top-k gate?* The answer to the first, on `simple_spread`, is yes at moderate N : dense attention communication beats MAPPO at $N=6$ with Cohen’s $d=1.06$ and a 95% bootstrap CI that excludes 0. The answer to the second is *not with the architecture we tested*: an end-to-end Gumbel-softmax gate over the $N-1$ destinations introduces seed-dependent variance that scales with N and overwhelms the 21–29% FLOPs reduction it actually buys. At $N=12$, TopK-TarMAC is significantly worse than MAPPO. The diagnostic — between-seed std of 20–40 for adaptive- k vs ~ 5 for dense and baseline — points to a clear next step: replace the unsupervised Gumbel gate with a Lagrangian-budgeted constraint on k_{eff} or on the message KL, so that sparsity is a controlled input rather than a noisy gradient. The reduce-to-baseline switch we expose in the implementation makes that experiment a one-flag change on top of the current code.

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